# RIVER VALLEY HIGH SCHOOL
## JC 2 PRELIMINARY EXAMINATION

| | |
|---|---|
| CANDIDATE NAME | |

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|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CENTRE NUMBER | S | | | | | CLASS | 21J | INDEX NUMBER | | | | |

**BIOLOGY** **9744/02**

Paper 2 Structured Questions **12 September 2022**

**2 hours**

Candidates answer on the Question Paper.

No Additional Materials are required.

**READ THESE INSTRUCTIONS FIRST**

Write your Centre number, index number and name in the spaces at the top of this page.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.
DO **NOT** WRITE ON ANY BARCODES.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [ ] at the end of each question or part question.

| For Examiner's Use | |
|---|---|
| **1** | |
| **2** | |
| **3** | |
| **4** | |
| **5** | |
| **6** | |
| **7** | |
| **8** | |
| **9** | |
| **10** | |
| **11** | |
| **Total** | |

Answer **all** questions.

**1** In the lung, epithelial cells have a thin layer of watery mucus on their surface to trap particles.

Fig. 1.1 shows a series of processes that occur at the cell surface membrane of an epithelial cell, involving a channel protein called the Cystic Fibrosis Transmembrane Conductance Regulator (CFTR).

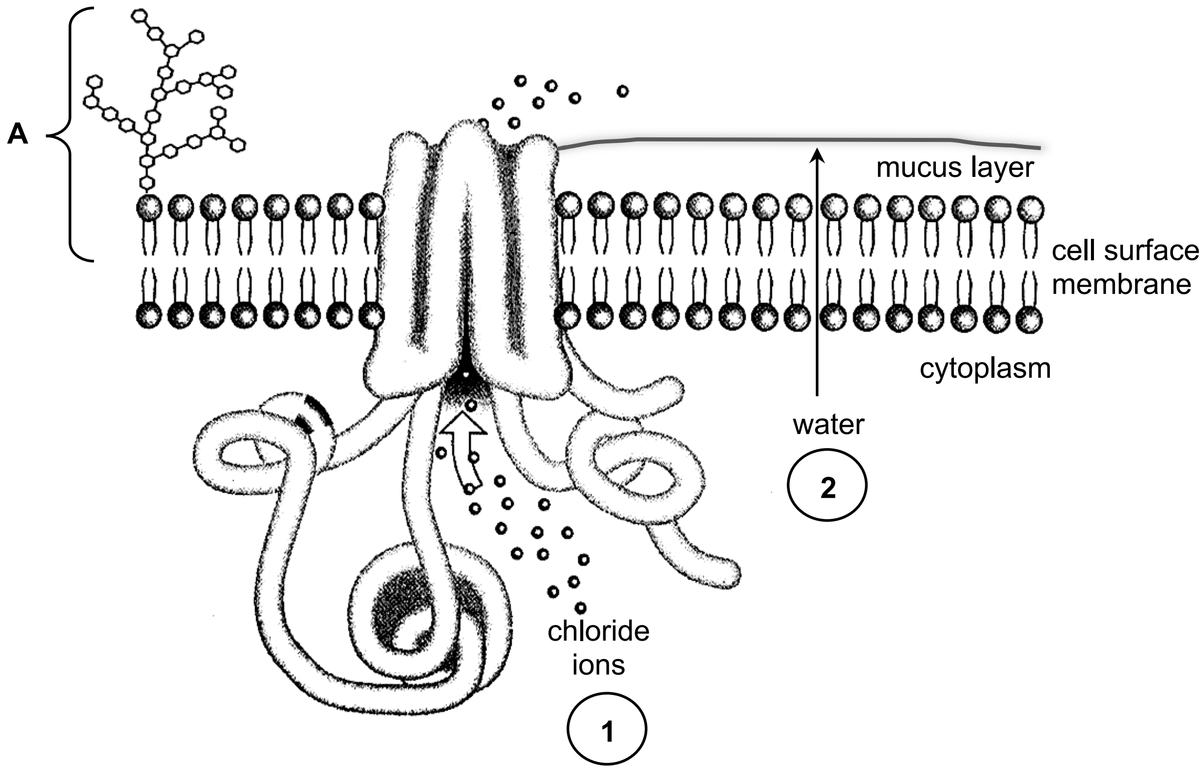



**Fig. 1.1**

**(a)** Identify the structure labelled **A** and state its role. [2]

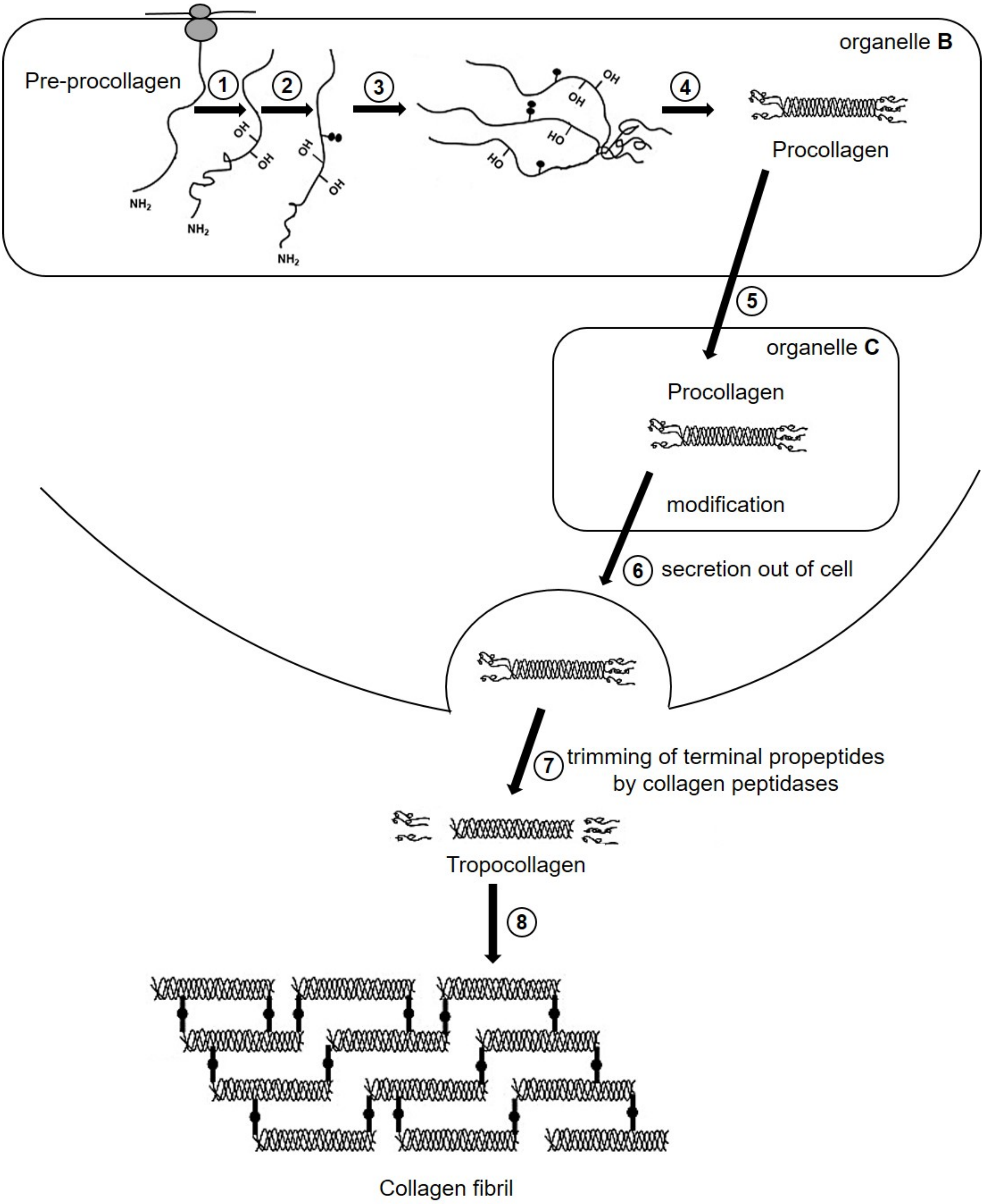
structure ______________________________

role ______________________________

**(b)** **(i)** Explain why chloride ions can only cross cell surface membranes via CFTR. [3]

**(ii)** Explain why absence of CFTR results in more viscous mucus. [2]

Fig. 1.2 shows the stages during the synthesis of collagen.




**key**

OH: hydroxyl group

: sugar

**Fig. 1.2**

**(c)** **(i)** Identify organelles **B** and **C** in Fig. 1.2. [2]

organelle **B** ______________________________

organelle **C** ______________________________

**(ii)** With reference to Fig. 1.2, describe how tropocollagen is synthesised from pre-procollagen. [3]

**(iii)** The feet of elephants are covered in padding to support the mass of the elephant. This padding is made up of a large number of cells surrounded by connective tissue containing many fibres of collagen.

Explain how **stage 8** contributes to the function of the padding. [2]

**(iv)** Suggest why collagen fibril is assembled outside the cell. [1]

[Total: 15]

**2** Glucoside hydrolase (GH) is an enzyme of important commercial use. GH can be immobilised in alginate beads to improve its physical and biochemical properties. A study was carried out to investigate the effect of pH on the activity of free GH and immobilised GH. Equal concentrations of free GH and immobilised GH were used.

Fig. 2.1 shows the results of this study.

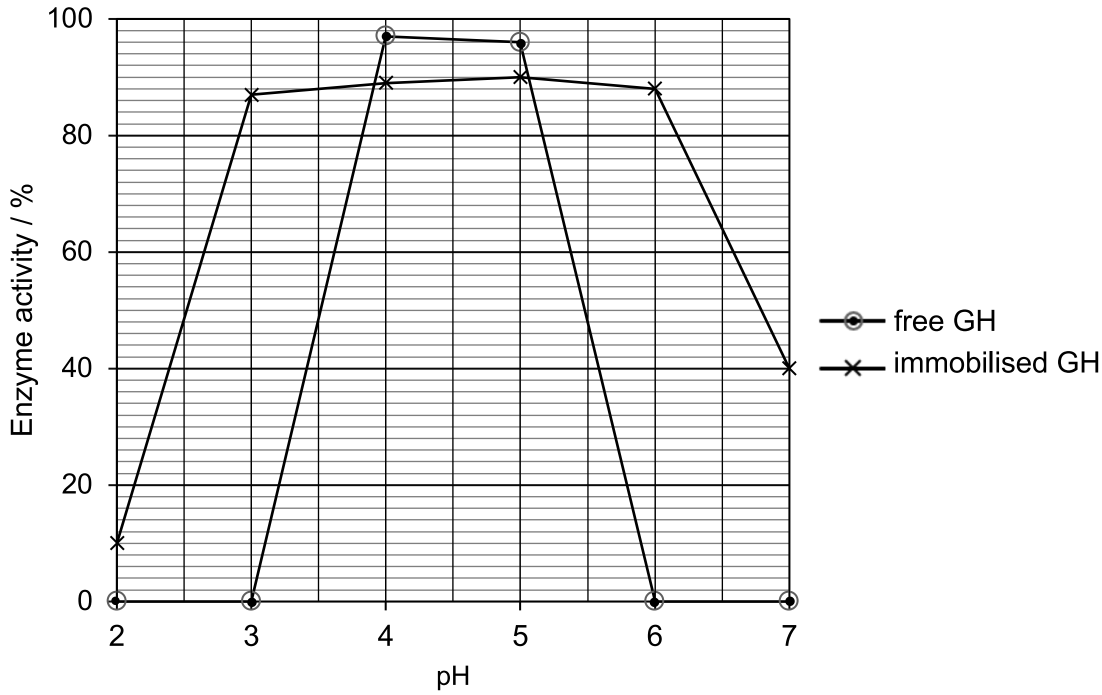



**Fig. 2.1**

**(a)** With reference to Fig. 2.1, describe **two** differences between the activity of free GH and immobilised GH.

Suggest an explanation for each of the two differences. [4]

difference 1 and explanation: ____________________

____________________

____________________

____________________

____________________

difference 2 and explanation: ____________________

____________________

____________________

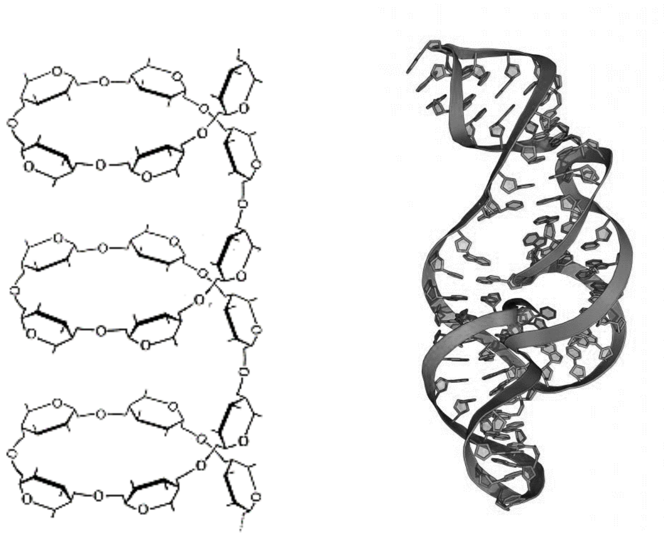
____________________

____________________

GH is an enzyme responsible for hydrolysing α-1,4-glycosidic bonds in carbohydrates, such as amylose and amylopectin.

**(b)** Using GH as an example, explain induced-fit hypothesis. [2]

Fig. 2.2 shows a molecule of amylose and a molecule of RNA (not drawn to scale).



**Amylose** **RNA**

**Fig. 2.2**

**(c)** Describe in what ways the helical structures in both amylose and RNA differ. [2]

[Total: 8]

**3** Fig. 3.1 shows a diagram of RNA polymerase during transcription of the β-globin gene.

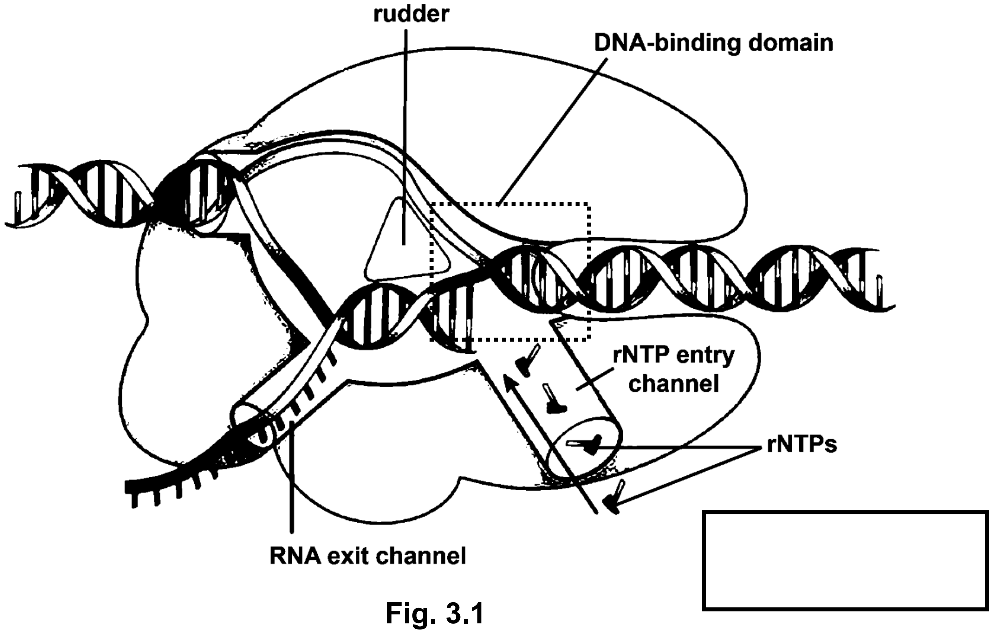



**Fig. 3.1**

**(a)** **(i)** In the box on Fig. 3.1, label with an arrow to show the direction of transcription. [1]

**(ii)** With reference to Fig. 3.1, describe how the structure of RNA polymerase allows it to perform its function. [2]

Actinomycin D and cycloheximide are drugs used in the treatment of chronic leukemia, and are involved in the inhibition of β-globin synthesis.

Fig. 3.2 shows the results obtained when each drug is added to immature red blood cells in separate experiments. The thickness and intensity of the bands indicate the amount of β-globin mRNA or β-globin protein present. Results have been adjusted to allow for direct comparison.

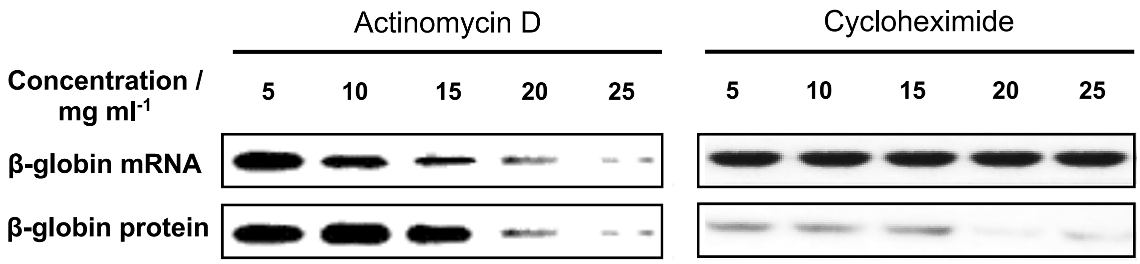



**Fig. 3.2**

**(b)** **(i)** State the drug that inhibits the following processes in β-globin synthesis. [1]

transcription

translation

**(ii)** Justify your answer in **(b)(i)**. [3]

**(c)** Describe **two** features of the genetic code. [2]

[Total: 9]

**4** Cancer is a genetic disease. Despite this, only around 10% of all cancers are inherited.

**(a)** Suggest **two** reasons why a cancer that occurs in a parent does not always occur in the offspring. [2]

Acute myeloid leukemia (AML) is a rare and extremely rapid developing cancer in human. AML results from aberrant proliferation and differentiation of mutated myeloid cells in the bone marrow. Large numbers of malfunctioning differentiated cells then result in symptoms such as fatigue, unusual bruising, and a high number of infections.

Patients with AML display mutations in many genes but mutations in *FLT3* and *DNMT3A* genes are most prevalent. FLT3 and DNMT3A are a type of receptor tyrosine kinase and DNA methyltransferase respectively.

Fig. 4.1 shows the development of AML in the bone marrow.

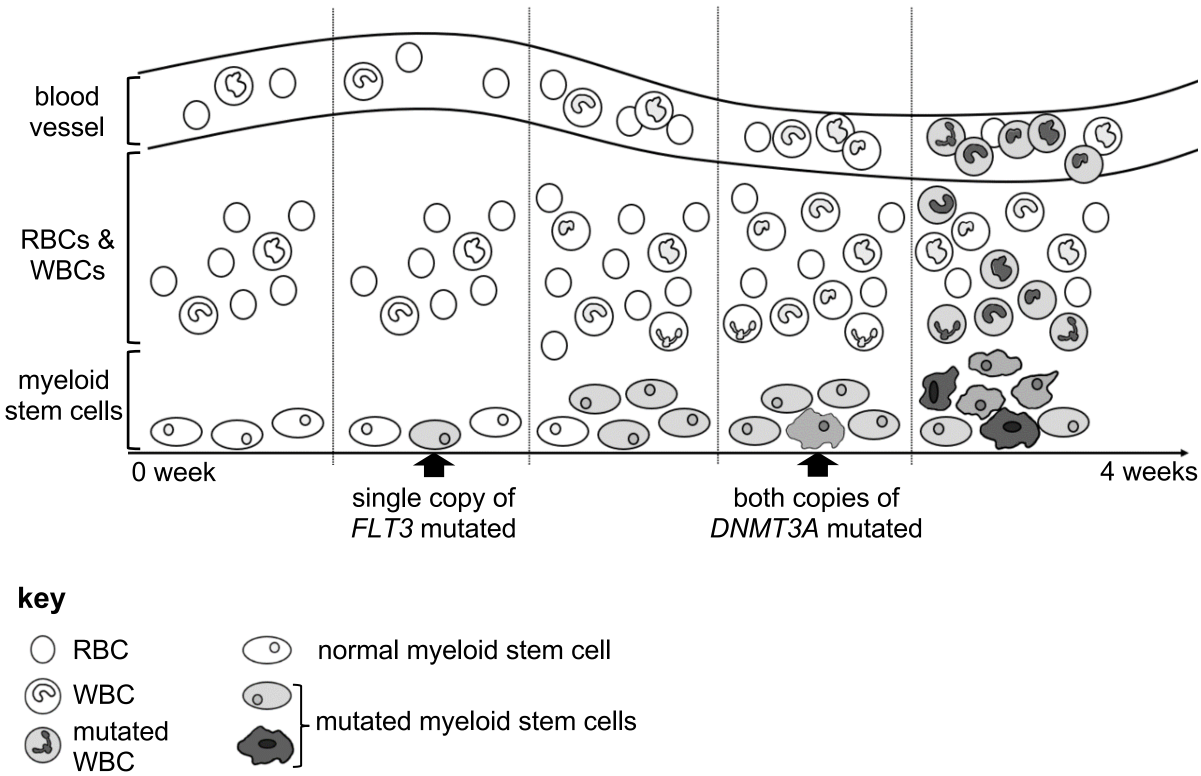
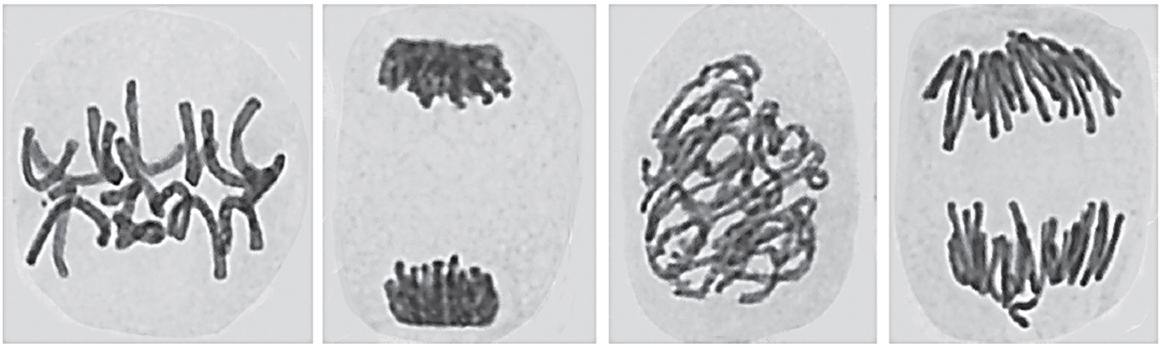



**Fig. 4.1**

**(b)** Outline how the mutations in myeloid stem cells result in the changes visible in Fig. 4.1 and the formation of multiple secondary sites of tumour development. [4]

Myeloid stem cells were obtained from a patient at 0 week. The photomicrographs below show the myeloid stem cells in various stages of nuclear division.

1 2 3 4



**(c)** **(i)** State which cell(s) contain twice as many chromosomes as a cell in resting phase from the same patient. [1]

______________________________________________

______________________________________________

**(ii)** With reference to cells 1, 2 and 4, describe in correct order the behaviour of chromosomes in these stages of nuclear division. [3]

______________________________________________

______________________________________________

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[Total: 10]

**5** Haemagglutinin (HA) protein of influenza virus is a homotrimeric protein complex. Each subunit is coded for by the *HA* gene present on one of its eight genomic segments.

Fig. 5.1 shows one of the subunits.

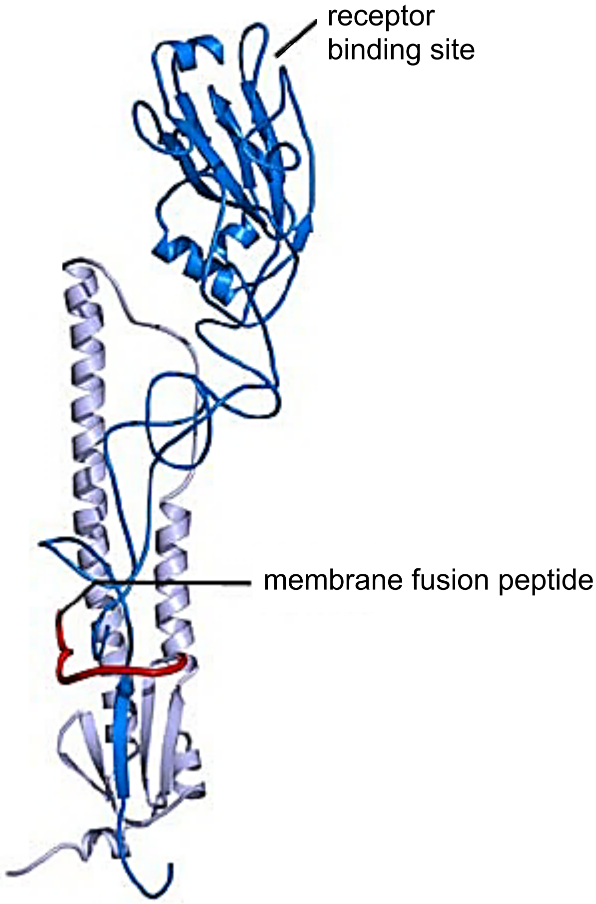


**Fig. 5.1**

**(a)** Outline how the identified features in Fig. 5.1 facilitate influenza infection. [3]

**(b)** Explain how influenza virus with eight genomic segments can result in a pandemic. [2]

Fig. 5.2 shows the number of deaths from influenza and the number of new cases of influenza from 1925 to 1991 in Country K.

An antiviral drug for influenza was introduced in Country K for widespread use in 1930. The antiviral drug specifically targets non-envelope viral proteins to reduce the influenza's ability to multiply.

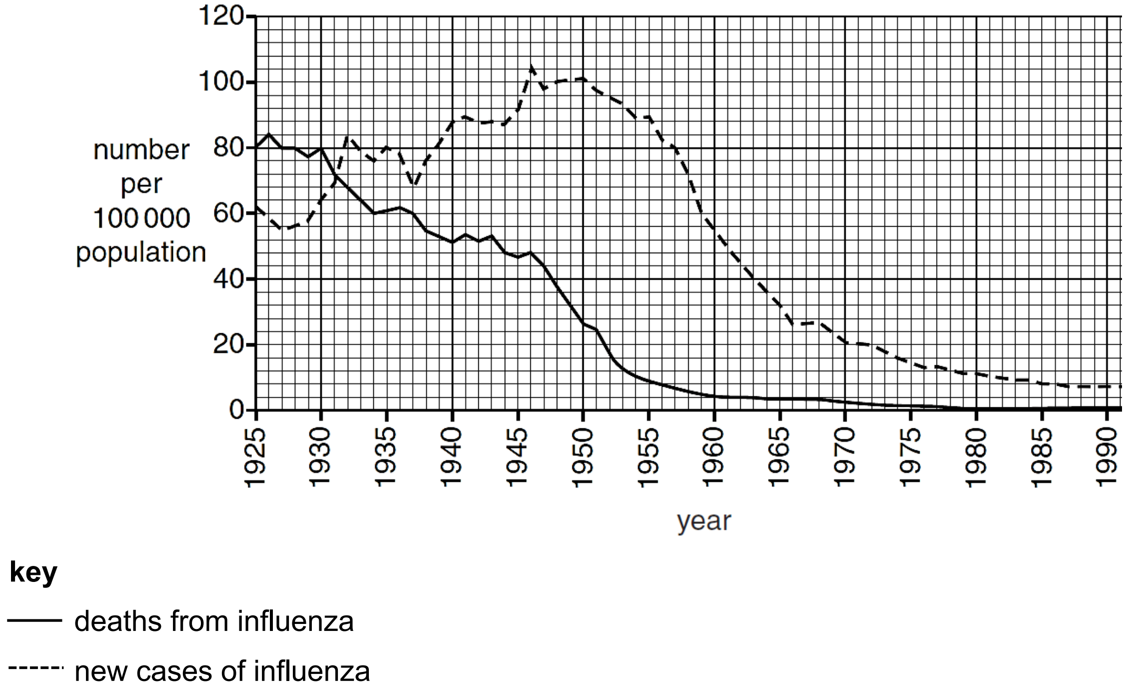


**Fig. 5.2**

**(c)** Using the information provided, account for the difference in the ten-year effect of the introduction of the antiviral drug on the number of deaths and number of new cases of influenza. [5]

[Total: 10]

**6** Gene expression in eukaryotic cells is regulated at multiple levels.

Fig. 6.1 shows the regulation of transferrin receptor (TfR) expression using Iron Response Element-Binding Proteins (IRE-BP). IRE-BP binds to IREs which are loop structures found on the mRNA.

TfRs are present on the cell surface and are involved in the uptake of extracellular iron.

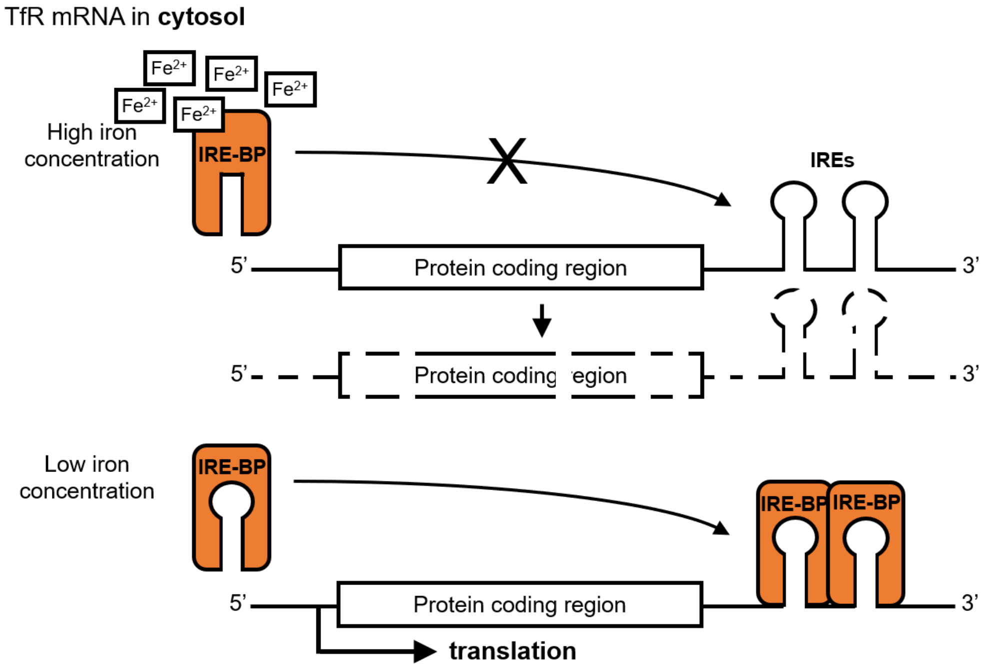


**Fig. 6.1**

**(a)** **(i)** State the level of regulation shown in Fig. 6.1. [1]

______________________________________________

**(ii)** Explain how a change in cytosolic iron concentration results in an increased uptake of extracellular iron. [4]

______________________________________________

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Ferritin proteins are involved in the storage of iron taken into the cell by TfRs. The regulation of ferritin expression is shown in Fig. 6.2.

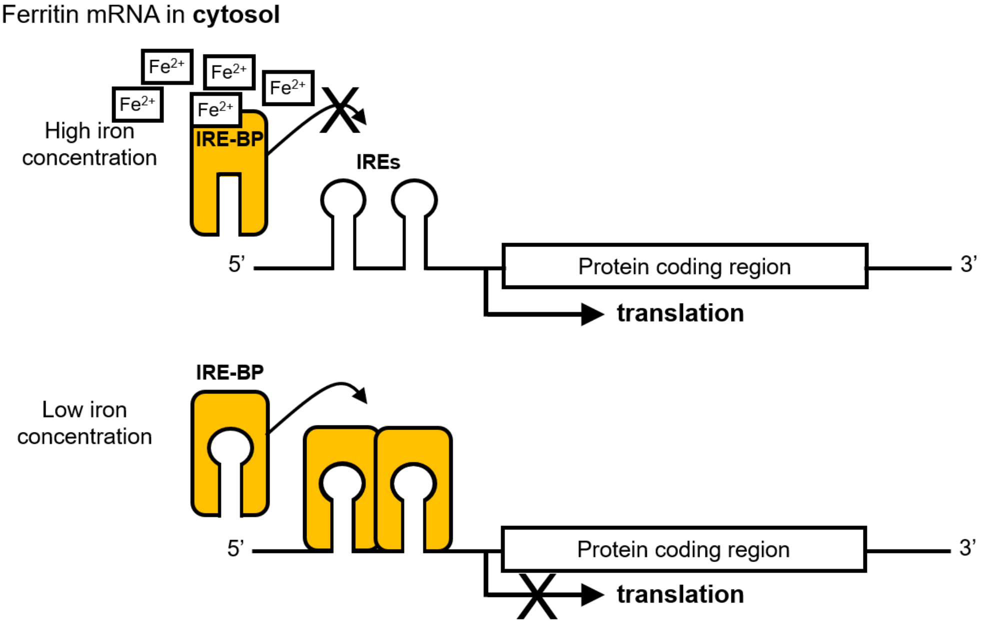



**Fig. 6.2**

**(b)** Outline two differences in the regulation of ferritin and TfR expression. [2]

At a different level of regulation, activators and repressors act on distal elements to control the rate of transcription.

**(c)** Describe how activators alter the expression of a gene. [3]

[Total: 10]

7 Dachshunds have three basic coat types: wire-, smooth- or long-haired. These are affected by two genes, **H/h** and **G/g**. The presence of **H** always results in wire hair. When long-haired dogs are crossed amongst themselves, they always produce long-haired puppies.

Two groups of dogs heterozygous for both genes are crossed and the results from this cross are shown in Table 7.1.

**Table 7.1**

| phenotype | number of offspring |
|---|---|
| wire-haired | 90 |
| smooth-haired | 22 |
| long-haired | 8 |

Fig. 7.1 shows the pedigree for inheritance of coat type in Dachshunds.

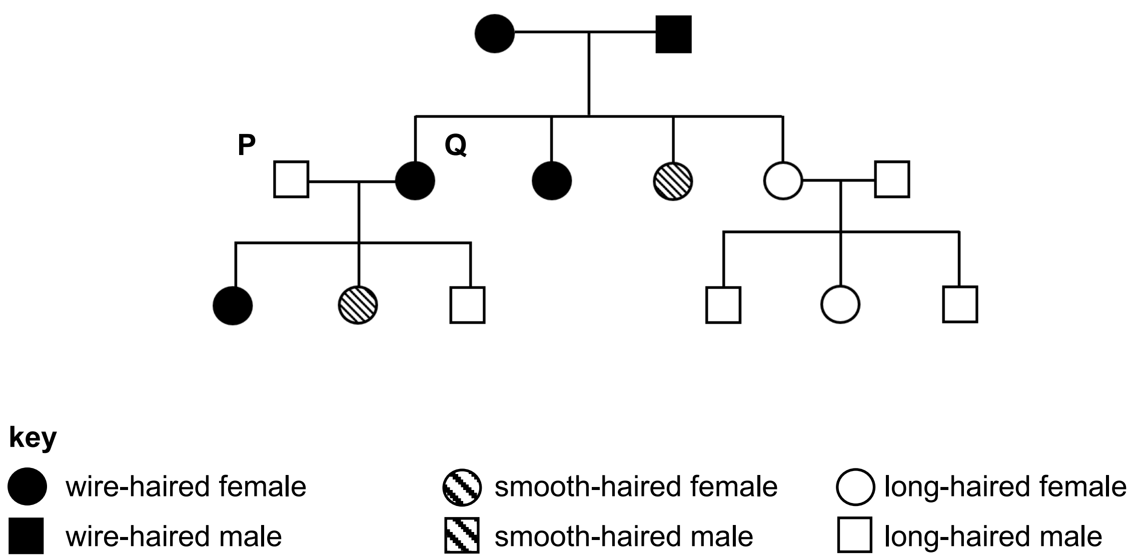



**Fig. 7.1**

**(a)** Draw a genetic diagram in the space below showing the cross between **P** and **Q**.

Use the symbols given and show all possible genotypes and phenotypes for the offspring of these parents. [4]

**(b)** In any genetic cross, the observed results are usually different from the expected results.

Suggest **one** reason why such a difference may occur, referring only to events after meiosis. [1]

Other than variation in coat type, dogs also display variation in their coat colours.

Table 7.2 shows the colour scale and number of dogs showing each of the phenotypes.

**Table 7.2**

| colour | | | | | | |
|---|---|---|---|---|---|---|
| colour scale | 1 | 2 | 3 | 4 | 5 | 6 |
| number of dogs | 114 | 216 | 305 | 350 | 288 | 133 |

**(c)** Distinguish between the two types of variation shown in coat type and coat colour in dogs. [3]

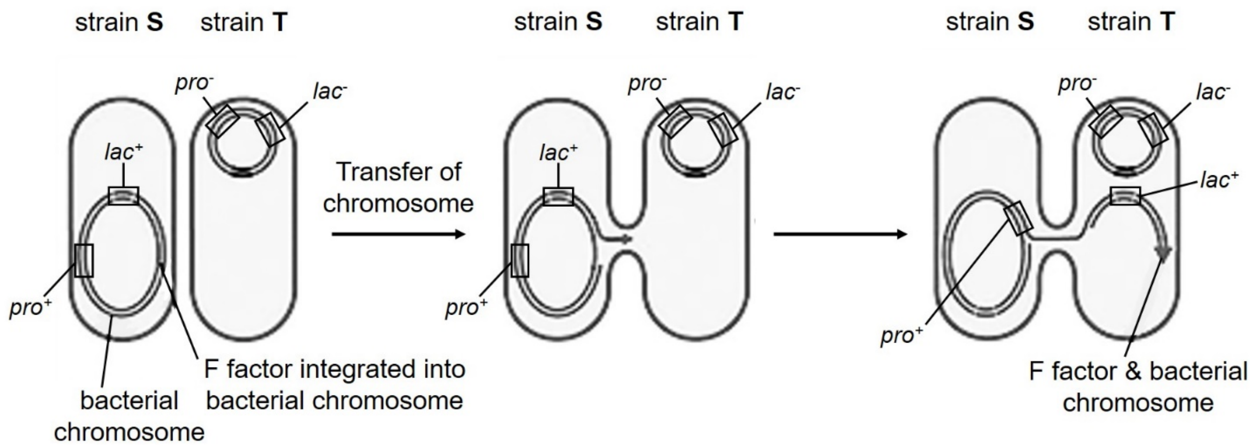
[Total: 8]

**8** **(a)** **(i)** Using the *trp* operon as an example, explain what is meant by negative gene regulation. [1]

**(ii)** Explain why a single base-pair insertion in the regulatory gene of the *trp* operon may lead to an over-production of tryptophan. [5]

An investigation was conducted to study gene transfer between two strains of *E. coli*, **S** and **T**. Bacterial cells of strain **S** are able to synthesise proline and metabolise lactose, while bacterial cells of strain **T** are unable to carry out both processes.

Fig. 8.1 shows the process of gene transfer during the investigation.




**key**

*pro*$^+$ : gene for proline synthesis present

*lac*$^+$ : gene for lactose metabolism present

*pro*$^-$ : gene for proline synthesis absent

*lac*$^-$ : gene for lactose metabolism absent

**Fig. 8.1**

**(b)** **(i)** With reference to Fig. 8.1, describe the gene transfer process between *E. coli* strains **S** and **T**. [3]

The two strains were mixed together in suspension in two separate experiments, for an incubation period of 10 minutes and 60 minutes respectively. At the end of the incubation, bacterial cells were then plated on an agar medium containing lactose and deficient in proline.

Table 8.1 shows the results of the two experiments.

**Table 8.1**

| *E. coli* strain | colonies present in nutrient agar | |
|---|---|---|
| | after 10 minutes incubation | after 60 minutes incubation |
| **S** | 94 | 95 |
| **T** | 42 | 93 |

**(ii)** Explain why there is a larger number of *E. coli* strain **T** colonies when the two strains were incubated together for 60 minutes as compared to 10 minutes. [2]

[Total: 11]

9 Fig. 9.1 shows the phylogenetic relationship of four different species of snapping shrimps obtained based on comparison of nucleotide sequence.

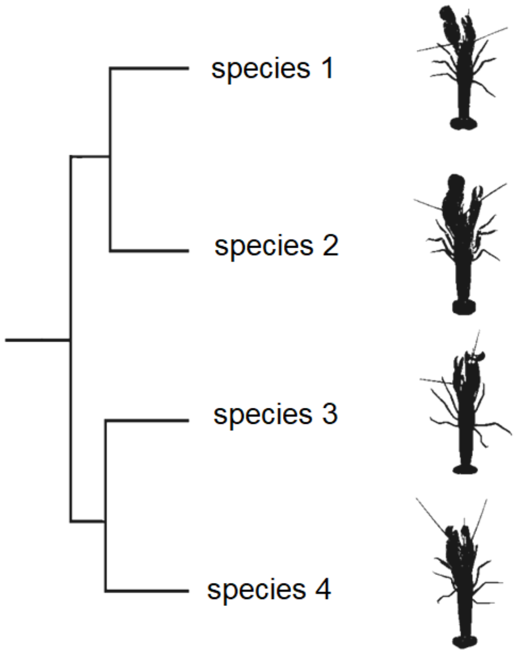



**Fig. 9.1**

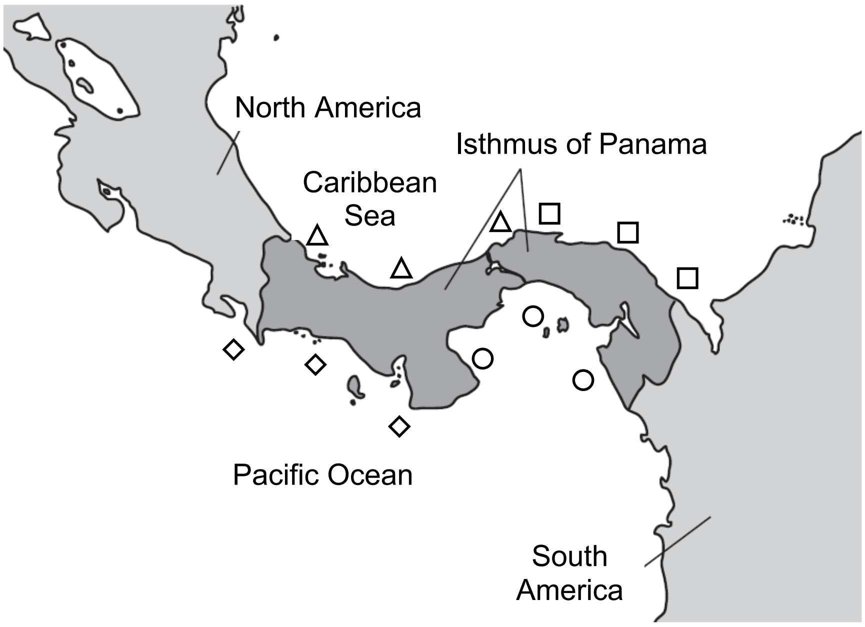
**(a)** Describe how molecular data was used to construct the phylogenetic tree. [2]

Twenty million years ago, there was a gap between the continents of North America and South America through which the waters of the Caribbean Sea and Pacific Ocean flowed freely.

The snapping shrimps lived in this area between North America and South America.

About 3 million years ago, volcanic activity and sedimentation formed a narrow strip of land, Isthmus of Panama, joining North America and South America.

Fig. 9.2 shows the current distribution these four species.




**key**

□ species 1 ○ species 3

△ species 2 ◇ species 4

**Fig. 9.2**

Using information from Fig. 9.1 and Fig. 9.2,

**(b)** **(i)** explain how biogeography and anatomical homology of snapping shrimps support Darwin's theory of evolution. [4]

**(ii)** suggest why breeding between species 1 and species 2 is possible. [1]

**(c)** Explain how fossil records can also be used as evidence for evolutionary change of species. [2]

[Total: 9]

**10** **(a)** Describe the role of T lymphocytes in the immune response against *Mycobacterium tuberculosis* infection. [3]

**(b)** Explain **two** ways in which immunoglobulin differ from penicillin in response to infections by pathogen. [2]

[Total: 5]

**11** El Niño phenomenon is characterised by increased temperatures in the oceans.

**(a)** Explain how El Niño affects the corals. [2]

**(b)** **(i)** Describe how **one** human activity increases global sea temperatures. [1]

**(ii)** Suggest how reduction in the diversity of coral reefs reduces biodiversity. [2]

[Total: 5]